

INM707 Detailed Experimental Plan

Self-Learning Snake Xenzia using Reinforcement Learning

This document presents the detailed experimental execution plan for the INM707 Deep Reinforcement Learning coursework project.

1. Research Questions

- 1 Can a tabular Q-learning agent learn an effective policy for Snake Xenzia?
- 2 How do reward shaping and exploration strategies affect learning performance?
- 3 Does DQN outperform tabular Q-learning in larger environments?
- 4 Do advanced DRL algorithms improve stability and performance compared to standard DQN?

2. Global Hypothesis

Deep Reinforcement Learning methods are expected to outperform classical Q-learning in terms of scalability, stability, and learning efficiency.

3. Experimental Pipeline

Environment → Q-Learning → DQN → Advanced DRL Extensions

4. Environment Development

Build a Snake Xenzia environment with movement mechanics, food spawning, collision detection, and Gym-style APIs.

Hypothesis

A compact state representation will improve convergence speed.

Expected Outcome

Stable and reproducible RL training is expected.

5. Tabular Q-Learning

Implement Q-learning with epsilon-greedy exploration and multiple hyperparameter experiments.

Hypothesis

The agent will learn basic survival and food-seeking behaviour.

Expected Outcome

Performance should improve gradually but degrade in large environments.

6. Deep Q-Network (DQN)

Replace the Q-table with a neural network using PyTorch with replay buffers and target networks.

Hypothesis

DQN will scale better than tabular Q-learning.

Expected Outcome

Higher rewards and better generalisation are expected.

7. Double DQN — Pritish

Implement Double DQN to reduce overestimation bias.

Hypothesis

Double DQN will improve stability.

Expected Outcome

More stable reward curves are expected.

8. PPO — Roberto

Implement PPO as a policy-gradient approach.

Hypothesis

PPO will demonstrate smooth policy learning.

Expected Outcome

Stable training and strong exploration behaviour are expected.

9. Hyperparameter Experiments

Experiment	Hypothesis	Expected Result
Exploration Rate	Slow epsilon decay improves learning	Better long-term policies
Reward Shaping	Dense rewards accelerate learning	Faster convergence
Environment Size	Large state spaces reduce performance	Poor scalability for tabular RL

10. Evaluation Metrics

Metric	Purpose
Average Reward	Overall performance
Maximum Score	Peak capability
Episode Length	Survival behaviour
Training Stability	Variance analysis
Convergence Speed	Learning efficiency

11. Conclusion

This project demonstrates a progression from classical reinforcement learning to advanced deep reinforcement learning approaches while clearly separating shared and individual contributions.